Exercise 1.4.1

Task 1

Suppose you take out a loan with a principal amount of #100,000. If the annual interest rate on the loan is 5% and you plan to repay it in 3 years, how much interest will you owe at the end of the loan term?

For this task write a function called `interest()` that accepts the 3 parameters as input arguments and calculates the amount of interest owed at the end of the loan term. The 3 input arguments should be in the given order:

- The principle loan amount
- The annual interest rate
- The number of years in which the loan should be repayed

Once you have written the function, call it with the provided parameters:

- Principal (P) = #100,000
- Rate (R) = 5% per annum
- Time (T) = 3 years

Solution

```
### CG T1
### Write your function here:


def interest(principal, rate, years):
  # Simple Interest = Principal * Rate * Time  (PRT)
  # Since rate is given as a percentage, i have to divide by 100
  return principal * (rate / 100) * years
```

```
### Run your function with the provided parameters here:

principal_amount = 100000
annual_rate = 5
loan_years = 3

interest_owed = interest(principal_amount, annual_rate, loan_years)
print(f"The interest owed at the end of the loan term is: #{interest_owed:.2f}"
```

```
The interest owed at the end of the loan term is: #15000.00
```

Task 2

Write a function called `celsius_to_fahrenheit()` that converts a temperature from degrees Celsius to degrees Fahrenheit.

You can use this formula:

$T_c * 1.8 + 32$

where T_c is the temperature in degrees celsius to be converted.

Once you have written the function, use it to convert the following temperatures from Celsius to Fahrenheit:

1. 100
2. 75
3. 120

Solution

```
### CG T2
### Write your function here:
def celsius_to_fahrenheit(celsius_temp):
  # using the formula: Tc * 1.8 + 32
  fahrenheit_temp = celsius_temp * 1.8 + 32
  return fahrenheit_temp
```

```
### Run your function with the provided parameters here:

# convert 100 degrees Celsius
temp1_c = 100
temp1_f = celsius_to_fahrenheit(temp1_c)
print(f"{temp1_c}°C is {temp1_f}°F")

# convert 75 degrees Celsius
temp2_c = 75
temp2_f = celsius_to_fahrenheit(temp2_c)
print(f"{temp2_c}°C is {temp2_f}°F")

# convert 120 degrees Celsius
temp3_c = 120
temp3_f = celsius_to_fahrenheit(temp3_c)
print(f"{temp3_c}°C is {temp3_f}°F")
```

```
100°C is 212.0°F
75°C is 167.0°F
120°C is 248.0°F
```